A Comprehensive Guide to Synthetic Data in Python: What, Why, and How — Part 1

5 min read · Mar 26, 2025

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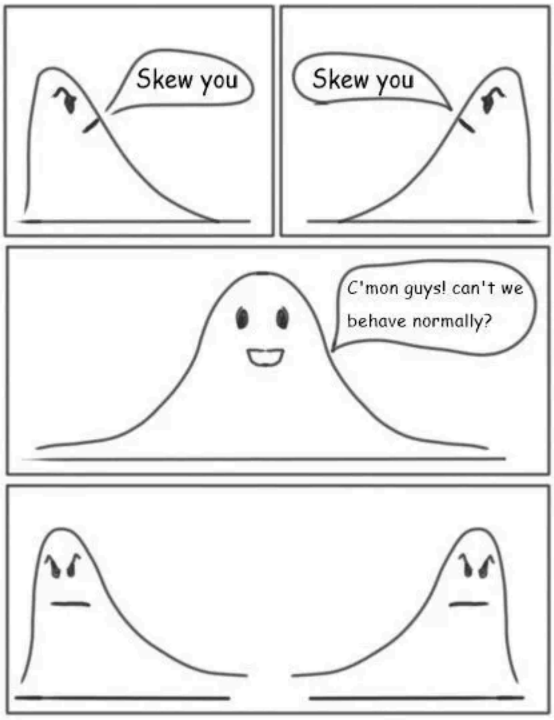
This is a two part article series. Part one of the article series will start by covering a primer on synthetic data. It is essential to get the background on the what and why of synthetic data before we delve into how this space is evolving as an AI enabler and a business product. Additionally, we discuss some considerations and challenges that come along with using synthetic data for AI/ data products. Part 2 of this article series will then discuss the evaluation methodology and methods to generate synthetic data with some examples in Python.

What and why?

Data can be complex, funny, mysterious, messy, pricey, wrong, biased, inadequate and many more things. Today's digital footprint and the amount of data generated each second is well known to both the consumers and the generators of those data. What still remains as a point of contention are the rights/ regulations/ rules and ownership of that data. Here is where the lines get blurry and we are now in an era where the large language models are already creating a storm while being built on data which was scraped from the entire internet. Who is to say what violation of privacy, ownership or bias may have taken place in this process?



Source : https://coolinfographics.com/news/2021/10/4/statistic-joke

In such a tricky and fast evolving space — synthetic data is trying to make a dent to resolve if not all but some of the issues. While the governments and regulatory bodies scramble around the world to develop data governance and responsible AI principles — synthetic data offers a more close, hands-on, practical solution to many of the challenges around data privacy, bias , inadequacy, etc. to name a few.

Synthetic data is basically artificially generated content which follows similar statistical properties of the original data across factors such as distributions, correlations, linearity/ non-linearity, etc. It could also be different from the original data after mitigation steps for issues such as bias or imbalance. This data could be tables, images, audio or text. The essence of this data is to mimic and be indistinguishable from real world data (*high fidelity*) yet not be related to any individual, transaction or real life data attribute in anyway (*privacy preserving*).

> *"Synthetic data is like a good politician: it can be used to support any argument"*

When it comes to practical use cases of synthetic data, what does it get down to?

Source : Author

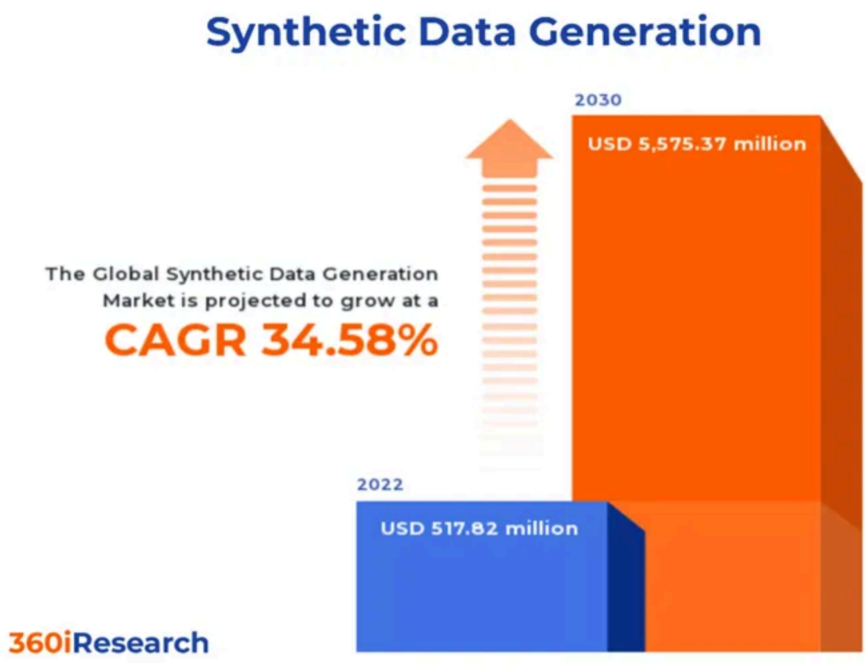
- New software builds could leverage synthetic data for testing purposes & new product builds could leverage simulation testing on synthetic data
- Synthetic data could augment your existing datasets within the organization and make it more mature and ready for AI uses cases

- Synthetic data can help the organizations tackle privacy and regulatory concerns around data which hinder most of the AI use cases and sharing of data to develop market models
- New start-ups and organizations can leverage the cheaper access to data which is synthetically generated versus buying data, where the data is not only expensive but markets are also at different maturities across continents
- Synthetic data curation can help address current issues of bias and quality, which can ensure the creation of more fairer and robust AI systems

How does the current landscape look?

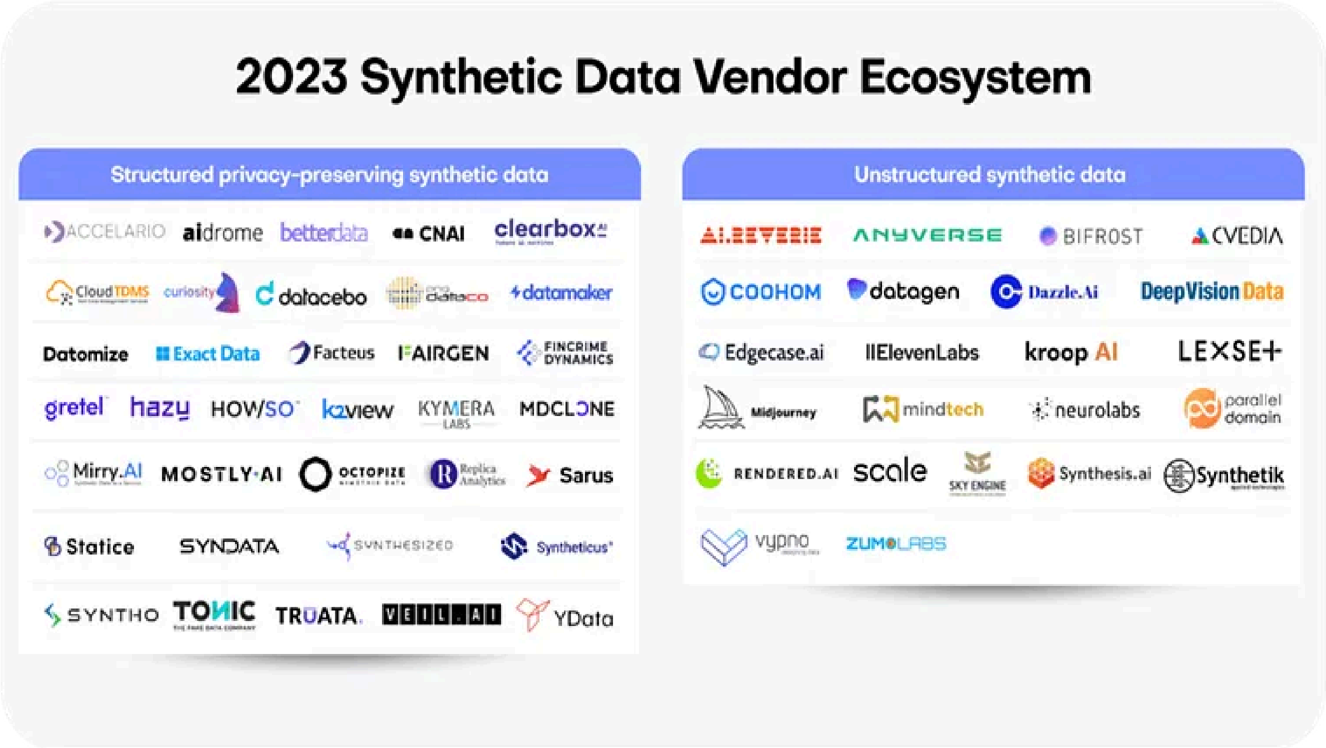
Across various surveys and studies, synthetic data industry shows a CAGR potential of more than 30% over the next decade.

> "T*his space is poised to grow with many new start ups and ventures coming in this space with the new generative era* "



Source : https://www.360iresearch.com/library/intelligence/synthetic-data-generation

As of today, there are many companies which have core product offering of context aware and high fidelity synthetic data.



Source : Source : https://mostly.ai/blog/synthetic-data-companies

Food for thought: Considerations and challenges

> **Open debate:** If algorithms learn the inherent statistical properties and attributes of the data and can generate new data which is identical to it, would that classify as a breach of privacy? Are the patterns, relations and statistical properties of data overall also under the purview of privacy and data protection? Will this be contested even if the data cannot be linked back to any individual, transaction or entity.

> **Quality & cost:** Data can have various considerations which are not just statistical but ethical and logical. Can algorithms capture these complexities well? How do the algorithms fare in a datasets which are high dimensional, contain complex multivariate interactions and are bound by endless business and domain logics & constraints? With these algorithm's high computation and high cost, does that challenge the relative ease of access to data where one buys the data externally.

> **Standardized frameworks:** This space is an active research topic still and there is still some way to go. There are no accepted standardized frameworks and methods which have matured and used actively across the AI community. There are varied approaches that have been mentioned in various papers. These frameworks need to be extensive to cover evaluations across multiple aspects which may just not be statistical. More of this is discussed in Part 2 of this series.

> **Market models and data sharing:** Could there be a data ecosystem where data sharing is enabled through privacy preserving synthetic datasets which together can foster open source development of robust AI solutions that have a social purpose?

What next?

In the second part of this article we will cover various methods and frameworks in Python to generate synthetic data:

- **Random and deterministic data generators:** Exploring packages like Faker, Sklearn, SymPy and Trumania
- **GANs** (General Adversarial Networks): The front runner in terms of approaches to generate high fidelity synthetic data that can capture all complexities and intricacies inherent in the data. We will explore methods like SDV framework in python and various methods that are part of its framework.
- **LLM for synthetic data:** The new kid on the block that can generate random and deterministic data in minutes with the help of prompts.

We will also cover evaluation frameworks that will encompass **Visual, domain, Machine learning-based** and **statistical metrics-based evaluations.**

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